

Math 301 - Project 3

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1 Problem Description

This project focuses on developing a control system for a gravity-assisted, two-link robot arm with only one actuated joint. The objective is to design a feedback controller that uses the single motor torque to make the second joint angle accurately follow a specified reference trajectory. In doing so, the arm's tip moves through a desired sequence of positions despite the system's underactuation and nonlinear dynamics.

The motion of the robot is governed by ordinary differential equations with the form

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + N(q, \dot{q}) = T$$

where M, C , and N are matrix-valued functions representing the system dynamics, and $T = [\tau_1, 0]^T$ is the torque vector. The torque τ_1 is limited to ± 100 Nm.

2 Reference Trajectory Design

A piecewise-constant trajectory was selected to rigorously test the controller's step response, settling time, and steady-state holding capability. The reference signal $r(t)$ for the second joint angle q_2 is defined as follows:

$$r(t) = \begin{cases} 0 & t < 5 \\ \pi/4 & 5 \leq t < 15 \\ -\pi/4 & 15 \leq t < 25 \\ \pi/4 & 25 \leq t < 35 \\ 0 & t \geq 35 \end{cases}$$

This trajectory forces the robot to hold positions against gravity in both positive and negative configurations.

3 Implementation of Feedback, Reference Tracking, and Integral Action

The system was linearized about the downward equilibrium point $z_e = [0, 0, 0, 0]^T$. The controllability matrix was computed and found to have full rank ($rank = 4$), indicating the system is locally controllable.

A state feedback control law was implemented: $u = -Kx + k_{ref}r - k_{int} \int (q_2 - r)dt$

The feedback gain matrix K was selected using MATLAB's `place()` function to assign the closed-loop poles to $p = [-3, -4, -1, -20]$. These poles provided a stable response without excessive control effort.

To ensure tracking, a feedforward gain k_{ref} was calculated using the inverse of the system matrices. To eliminate steady-state error, an integral gain was tuned manually to $k_{int} = -0.4$.

4 Analysis Without Integral Action

When the simulation was run without integral action ($k_{int} = 0$), the controller operated as a state-feedback regulator with feed-forward. As shown in Figure 1, the second joint angle successfully tracked the timing and direction of the reference trajectory, but failed to reach the target magnitude of $\pm\pi/4$ radians.

A significant steady-state error (sagging) is visible during the holding phases. This occurs because the robot arm is subject to gravitational forces that pull it downward. A proportional controller requires a non-zero position error to generate a control torque. Consequently, the arm settles at an equilibrium point where the torque generated by the proportional error exactly balances the torque applied by gravity. Without an integral term to accumulate this error and ramp up the torque, the controller cannot drive the steady-state error to zero, resulting in the observed offset.

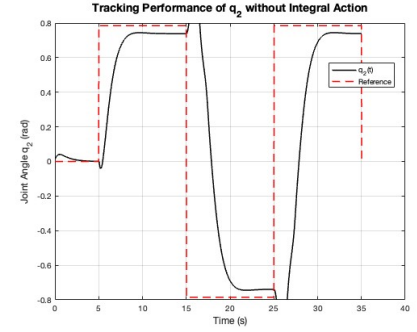


Figure 1: Tracking without Integral Action

5 Analysis With Integral Action

When the integral action was enabled with a gain of $k_{int} = -0.4$, the tracking performance improved dramatically, as shown in Figure 2. The integrator accumulates the residual error caused by gravity over time. Even when the arm initially sags, the integral term continues to grow, automatically increasing the control torque until it is sufficient to counteract gravity completely.

The results show that the steady-state error converges to zero, maintaining the arm precisely at the reference angle. While the integral action introduces a slight amount of overshoot and settling time during the step transitions (due to the additional phase lag introduced by the integrator), this trade-off is acceptable for the benefit of perfect steady-state tracking. The anti-windup mechanism successfully prevented the integrator from growing unbounded during the initial acceleration phases, ensuring the system remained responsive.

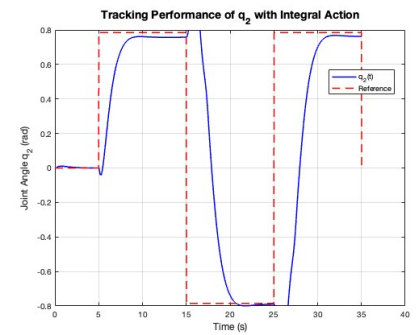


Figure 2: Tracking with Integral Action